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Can Ancient Indigenous Technology Help Save BC's Salmon?

After an earthquake exposed thousands of mysterious wooden stakes off Vancouver Island, researchers spent over a decade figuring out what they mean

BY BRIAN PAYTON

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Photo courtesy of Nancy Greene

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title as the most powerful onshore quake ever recorded in Canada. In nearby communities, brick walls fell and three-quarters of all chimneys collapsed. Two casualties were recorded that day: one man died of heart failure and another drowned after his dinghy was overturned by a wave generated when a piece of land gave way and thundered into the sea. For a while, that seemed like the end of the story. But, over time, the changes wrought by the

quake revealed a mystery that had lain hidden for generations—long enough to be forgotten.

Twenty-two kilometres from the quake's epicentre, locals started noticing wooden stakes appearing in the intertidal zone of Comox Harbour, on the east side of Vancouver Island. They ranged in size from the width of an adult's thumb to the width of an arm, but they stuck out little more than ankle high from the sand and mud. Locals pondered the mystery; many assumed they were the leavings of some relatively recent industrial activity or a fishing scheme abandoned by immigrants from Japan.

In 2002, Nancy Greene, then an undergraduate anthropology student, walked among the barnacle-encrusted stakes and thought she'd found a fascinating subject for her senior project at Malaspina College (now Vancouver Island University). She had lived in the area since 1978, had raised her children there, and was up for a new challenge. Little did she know, it would consume countless hours, span more than a decade, and eventually reveal the largest unstudied archaeological feature yet found on the Pacific Northwest coast—one that

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ON THE EASTERN slopes of the Beaufort Range, rain and meltwater flow down the Puntledge and Tsolum Rivers and converge in the Courtenay River before reaching Comox Harbour. These sheltered waters are part of the Salish Sea, which stretches from British Columbia's Inside Passage down to Washington state's Puget Sound. People have been living off the bounty of this marine environment ever since they began arriving in the region, near the end of the last ice age, about 13,000 years ago. Comox Harbour lies within the protected waters of a broad, gently sloping estuary that covers an area of 9.6 square kilometres, slightly bigger than San Francisco's Golden Gate Park. It is the traditional territory of speakers of the extinct Pentlatch language, whose descendants form part of the 342-member K'ómoks First Nation, which, along with the communities of Courtenay and Comox, now surrounds the harbour.

There had always been a few bits of wood poking up through the sand and mud of Comox Harbour, but after the quake of 1946, thousands of stakes emerged across vast stretches of the intertidal zone. This was likely a result of liquefaction, a phenomenon in which shaking reduces the strength of the sediment and leads to erosion. Subsequent periods of dredging near the mouth of the river may have also contributed to the process. It was clear that the stakes formed patterns, but just what those patterns represented was a puzzle until quite recently. In her interviews with members of local Indigenous communities, Greene found only one clue: a K'ómoks Elder said her

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what they were, they didn't seem to know. What was well known were the frequent battles that took place in the harbour before colonization. Those foolish enough to attempt a raid on the people living here faced harsh consequences. "What we did with people like that was chop their heads off, put them on a spear, toss them in the sand, and leave them as a reminder for other people not to come."

Frank clearly relishes relaying the tale, a testament to the abundance of salmon and the tenacity of the people protecting their claim to Comox Harbour. Now, as the history of the stakes is becoming known, he says they are a source of pride in his community.

Uncovering that history required hands-on research. In 2003, after surveying the entire estuary, Greene pulled on her gumboots and set out with pin flags and a laser theodolite to take geolocations of stakes across an area encompassing a total of approximately thirty hectares. She enlisted a team of volunteers including her husband, retired geologist David McGee, to help find and mark the stakes while trying to outrun the incoming tide. Because not all tides are created equal, she had to account for variations in how much area a tide exposed, available light, and weather. After months of reconnaissance, then weeks of recording geolocations, she recalls that first moment seeing the information they had collected displayed on a computer screen. Suddenly, those individual nubs of Douglas fir and western red cedar became 900 little black points on a field of white—like a photographic negative of stars in the nighttime sky. Patterns began to emerge and repeat. It took months of analysis, she says, before she began to realize what they represented—the remains of an immense, highly coordinated, sophisticated fish trap system, the largest such system discovered in North America, if not the world.

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or spawning grounds farther upstream swim in with the tide and can't escape. The ancient technology relied on a deep and intimate knowledge of local fish behaviour.

Evidence suggests that complex hunter-gatherer cultures around the world invented fish traps independently, at different places and times. Unlike wooden stakes, rock assemblages used in other fish weirs are difficult to date, but radiocarbon dating of adjacent middens (piles of fish bones and shells) offers a kind of proxy. Some of the oldest confirmed traps in North America are on mainland British Columbia, at the mouth of the Fraser River (4,500 to 5,280 years old) and in Maine (5,770 years old). The oldest-known fish traps, between 9,000 and 7,000 years old, were found in northern Europe. But the technology is probably far older. A line of stones found on the shore of an ancient lake in the Kenya Rift is reminiscent of the fish weirs used by the local people in modern times. It dates to the time of *Homo erectus*, or at least 490,000 years ago. If this was indeed a weir, it would mean the technology predates modern humans.

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The scale and complexity of the fish weirs found in Comox Harbour are staggering. Multiple traps were likely in use at the same time and, collectively, would catch immense quantities of fish. Over the course of their research, published in the *Canadian Journal of Archaeology* in 2015, Greene and her team recorded the positions of 13,602 stakes. Radiocarbon dating of fifty-seven stakes revealed ages ranging from 1,300 to just over 100 years old. Greene, now a research archaeologist, conservatively estimates that there are approximately 150,000 to 200,000 stakes in the harbour, which represent the remains of more than 300 fish traps. She knows of no other site approaching this scale of stake density.

In Comox Harbour, the patterns of the stake alignments reveal two distinct designs: one heart shaped and one chevron shaped. In both designs, removable lattice panels were likely lashed to the stakes to act as fences designed to lead the fish into the traps during high tide. When migrating fish encountered the barrier, they were directed into an opening at the crease between either the lobes of the heart or the wings of the chevron. As the tide receded, the fish inside the trap were stranded. The heart-shaped design mirrors historical fish weirs found in other sites along the Pacific Northwest coast, the east coast of

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The stakes from the heart-shaped traps correspond with the earlier dates returned from radiocarbon dating. They range in age from 1,240 years old to a little over 840 years old. Because of their proximity to nearby middens and the preponderance of herring bones in those middens, Greene suggests that the people of Comox Harbour used heart-shaped traps to catch herring. They built, operated, and maintained those traps during a prolonged era of warm temperatures and frequent droughts—an era that was coming to an end.

On the east coast of Vancouver Island, there was a marked increase in precipitation around 850 years ago. As the air got cooler and ocean temperatures dropped, fish ranges shifted. The archaeological record reflects these changes. After using and maintaining the heart-shaped traps for over four centuries, locals abruptly replaced them with the chevron-shaped design. Greene found no evidence of a period of trial and error. Knowledge of this new design probably already resided within the local population, or they quickly obtained it. “There were heart-shaped traps, and then there were chevron-shaped traps,” she says. “There were no traps in between.” It was a rapid technological adaptation to an altered climate.

The new chevron-shaped traps, which worked on the same principle of corralling schooling fish into a holding pen, were designed to catch much larger fish—up to thirty times the mass of herring. Locals built the traps to take advantage of a species multiplying exponentially in the cooler temperatures, a species that would come to support the very foundation, stability, and fluorescence of culture in Comox Harbour and all along the Pacific Northwest coast—a complex and sophisticated society that did not rely on agriculture. For the next five centuries, the people of Comox Harbour expanded, rebuilt, and maintained those traps for catching salmon.

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from the Pacific Northwest coast include some with handles and others with ergonomic thumb and finger grips etched into the stone. They repeated the process dozens of times, likely over numerous tide cycles, in order to create a single chevron-shaped salmon trap. Once the stakes were secure and the lattice panels were lashed in place—but before any salmon were taken—tradition dictates that the people would pay respect.

At the water's edge, a shaman would stand on a platform with his face painted red and eagle down in his hair—a symbol of peace and welcome. He would shake his ceremonial rattle and sing, then head out in his canoe. He would harpoon several salmon and put aside the first one he caught. The entire community would stand on the beach and watch, anticipating his return. When he came ashore, he would sing to and honour the first salmon by sprinkling it with eagle down. Once it was cooked and the feast complete, fishing could begin.

As salmon arrived in the harbour, in search of the freshwater surge from the Courtenay River and the spawning sites upstream, some encountered wooden panels that formed a barrier, forcing them through the narrow opening of a trap. One by one, the salmon followed one another inside, where they found themselves directed back along the wings of the pen, unable to escape.

During the salmon run, numerous fish traps would be in operation around the harbour. Men of high status or lineage probably controlled access to the traps. Traditionally, in cultures along the Pacific Northwest coast, men were responsible for catching fish. Women and young children most often processed fish; enslaved people, who were considered genderless, were also likely given this task. The traps worked day and night, in concert with the tide, until either

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the panels were removed, the rest of the fish could easily pass between the bare stakes to spawn in the rivers and streams beyond. An example of just such a panel, nearly six metres long and radiocarbon dated to the late fourteenth century, was discovered in Comox Harbour. The traps were highly consistent in form and were likely built using standardized units of measurement. One series of three linked traps, which may have been in use at the same time, stretched over a distance of more than three football fields (320 metres). The traps ensured that both the fish and the fishery thrived.

Greene suspects that the fishery in and around Comox Harbour would have supported a high population density. She believes the enormous number of fish caught and processed here would go to feeding not only the locals over the coming winter months: they likely traded fish up and down the coast and across the Salish Sea. Prior to the smallpox epidemic of 1862, there were about 30,000 Indigenous people living along the coast of British Columbia's Inside Passage. The fishery at Comox Harbour may have been the centre of cultural activity in the northern Salish Sea for at least 1,300 years.

Deidre Cullon, an archaeologist and adjunct professor in the geography department at Vancouver Island University, works for the Laich-Kwil-Tach Treaty Society. She has studied Pacific Northwest fish traps and wrote her doctoral dissertation on the relationship between Pacific Northwest peoples and salmon. "What I find," she says, "is that, the more we do and the more we learn, the more questions we have."

Cullon, like Greene, found it challenging to obtain any information about historical fish traps in the Indigenous communities she surveyed. Why has the

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knowledge lost, but entire communities were abandoned. Lost, too, was the need for a high-production fishery—there were far fewer mouths to feed.

“And then, right on the heels of that, the Canadian government chose to support commercial fishing for canneries,” Cullon explains. The government declared the traps illegal and sent their fisheries officers to destroy them. This was followed by the advent of the notorious residential school system, in which Indigenous children were removed from their families by the government and religious institutions and taken to far-off boarding schools, effectively separating them from their communities, languages, and cultures. This resulted in a profound disruption in the transfer of traditional knowledge, including the purpose and use of fish traps.

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As the tide recedes in the harbour today, remains of the stakes poke out of the estuary. Photo courtesy of Nancy Greene.

Although the ways and means of fishing changed, salmon retained their place at the heart of Indigenous society on the Pacific Northwest coast. Among many First Nations on this coast, it was taboo to toss salmon remains on a rubbish heap, as was done with herring and shellfish. People released the remains of salmon into the sea out of respect for what they considered nonhuman kin.

“The ocean was the water of life,” Cullon explains. “It had resurrection properties that allowed them to be reincarnated so that they can then return to the human world the following year.” In the Indigenous belief system, this respect and these traditions ensured the salmon’s return.

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future bring?

ON THE PACIFIC Northwest coast and around the world, change is underway again. On a bright summer day last year, a fisher hauled in evidence little more than eighty kilometres south of Comox Harbour. He was fishing for salmon but described his catch as “a metre long and all muscle and all teeth.” It was a Pacific barracuda, an aggressive predatory species common in the subtropical waters off Baja California, over 2,000 kilometres to the south. William Cheung, the Canada Research Chair in ocean sustainability and global change at the University of British Columbia, says that warmer-water fish, such as barracuda and ocean sunfish, are arriving in local waters with increasing frequency. He predicts a future in which sardines, a fish more associated with southern California, will become common on Canada’s west coast.

Cheung’s research also opens a window to the past. He can corroborate the shift in ocean surface temperatures approximately 850 years ago, temperatures that favoured salmon. And, now, he sees another shift underway. After centuries of relative stability, ocean surface temperatures will likely continue to rise on the coast of British Columbia over the next thirty years. His projections suggest that this warming will bring a 30 percent decline in sockeye salmon, but that’s only part of the story. Episodic marine heatwave events, such as the Blob, will exacerbate this baseline temperature increase—doubling the impact on fish like salmon.

Cheung says that the temperature increases he’s seeing now are resulting in changes that are beyond what people have experienced before. He’s concerned that adapting to those changes will be less straightforward in the future. What’s

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ensure the sustainability of natural systems. They organized their society around it. Today, as climate change accelerates and we continue the exploitation of global fish stocks to or beyond their capacity, modern society is leaving evidence of our commercial fishing philosophy in intertidal zones, on beaches, and adrift on and littering the bottom of the sea—much of it plastic. But, on British Columbia's central coast, just north of Vancouver Island, the Heiltsuk Nation is looking back to a traditional technology to help safeguard the future of its fishery.

William Housty, conservation manager for the Heiltsuk Integrated Resource Management Department, says evidence of ancient stone fish traps and cedar stake fish weirs is found throughout Heiltsuk territory.

“It's not like throwing a net in the water and catching every salmon that's swimming by,” he says of the old traps and weirs. They represent what he calls a brilliant technological approach because they were adapted on a creek-by-creek basis, which allowed for intimate knowledge and management focused on sustainability. Now, he says, the technology has proven to be invaluable for research.

Today, biologists commonly use weirs for monitoring fishery health, but the technology is rarely used in the Indigenous territories where it evolved. In 2013, the Heiltsuk Nation built a fish weir, based on a traditional design, on the Koeye River, an important salmon-bearing stream. It has allowed local people to identify, tag, and release salmon; to understand critical relationships between rates of salmon survival and spawning; and to monitor stream temperature fluctuations—in short, to assess the health of the ecological system.

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well that, in previous times, salmon were the main staple of our ancestors.”

The salmon caught in the new Heiltsuk weir are not yet used for food or ceremonial purposes. That will happen only once local managers are confident that sustainability objectives have been met. Housty looks forward to that day. When it arrives, he says, the first fish taken will be welcomed with honour and respect.

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About Brian Payton (View All Articles)

Brian Payton is the award-winning author of *Shadow of the Bear*, *The Ice Passage*, and the national bestselling novel *The Wind Is Not a River*. His nonfiction has appeared in the *New York Times*, the *Boston Globe*, and *Canadian Geographic*. He lives on Vancouver Island, BC.

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